



ENGINEERS LEVEL · SEMESTER 4

Data Detective

Student Workbook

Load data, compute the stats, visualize it — and reveal an insight the data supports.

Name: _____

THEORY

Finding The Story In Data

Load a **dataset** into a **list**, loop to compute stats — sum, len, **mean**, max/min — then **visualize** it to reveal a **trend**. An **insight** is a claim the data actually supports. Careful: a pattern is not always a cause (correlation is not causation).

Discuss: what real question could you answer with a small dataset, and what stat would help answer it?

Trace it

Read the Python and give the mean.

```
data = [4, 8, 6]
```

```
mean = sum(data) / len(data)
```

What is mean? _____

Code we use:

`list []``for (loop)``sum / len``max / min``mean``chart`

Words we are learning

- **Dataset** — a collection of data values to analyze.
- **List** — how the data is held in Python.
- **Mean** — the average: `sum(data)/len(data)`.
- **Trend** — a pattern over time or category.
- **Visualization** — a chart that makes the pattern clear.

Name: _____ Date: _____

Lists & Loops Over Data

Data lives in a **list**; a loop processes it. Trace and total.

```
data = [3, 7, 2, 8]
```

```
total = 0
```

```
for x in data: total = total + x
```

What is total? _____ How many values (len)? _____

Name: _____ Date: _____

Compute The Statistics

Compute basic **stats** for this dataset by hand.

```
scores = [10, 6, 8, 8, 3]
```

sum = ____ **len** = ____ **mean** = ____ **max** = ____ **min** = ____

Name: _____ **Date:** _____

Read A Chart

Interpret a simple chart and state the **trend**. Then think critically.

A chart shows library visits rising each month, Jan-Apr.

What is the trend?

Does rising visits prove the new posters caused it? Why or why not?

Name: _____ **Date:** _____

Plan My Investigation

Plan a data investigation from question to insight.

My question:

My dataset + the stat I'll compute:

The chart I'll make, and the insight I expect:

Name: _____ **Date:** _____

PROJECT

Data Detective — Plan

Analyze a real (small) dataset in Python: load it into a list, compute a statistic, make a visualization, and state one insight the data supports — noting a limitation.

1. My question and dataset:

2. Stat to compute:

```
mean = sum(data)/len(data)
```

3. Visualization type:

4. My insight (and one limitation):